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Grade:

### Fundamentals of theory of computation I. – sample

0-23: grade 1, 24-31: grade 2, 32-39: grade 3, 40-47: grade 4, 48-60: grade 5

1. True or false? Write the first letter of your answer (T or F) into the box. (20x1 pont)

- ☐  $ab$  is a suffix of  $abbaba$ .
- ☐  $L\{\varepsilon\} = L$  holds for every language  $L \subseteq \{a, b\}^*$ .
- ☐ Every language that can be generated by a linear grammar can be described by a regular expression.
- ☐ The regular expression  $cc^*$  describes the language  $\{c^{2n} \mid n \geq 0\}$ .
- ☐ Every language that can be described by a regular expression can be generated by a type 3 normal form grammar.
- ☐ Let  $A = (Q, \Sigma, \delta, Q_0, F)$  be a nondeterministic finite automaton. Then there is a regular expression  $R$  such that  $L(R) = L(A)$  holds.
- ☐  $G = (\{S, X\}, \{c, d\}, \{S \rightarrow cX, S \rightarrow \varepsilon, X \rightarrow ddX, X \rightarrow cd\}, S)$  is a grammar of type 3 normal form.
- ☐ Let  $G = (N, \Sigma, P, S)$  be a length-nondecreasing grammar. If  $u \rightarrow v \in P$  and  $u \neq S$  hold, then  $|v| \geq 1$ .
- ☐ If all rules of a grammar  $G = (N, \Sigma, P, S)$  is of the form  $A \rightarrow BC$ ,  $AB \rightarrow AC$ ,  $BA \rightarrow CA$ ,  $A \rightarrow a$  or  $A \rightarrow \varepsilon$  ( $A, B, C \in N, a \in \Sigma$ ) then  $G$  is a grammar of Kuroda normal form.
- ☐  $\{u \in \{a, b\}^* \mid \text{the number of } b\text{'s is odd in } u\}$  is a context-free language.
- ☐ The membership problem of type 2 grammars is decidable in polynomial time.
- ☐ If  $B \in N$  is a reachable nonterminal of a context-free grammar  $G = (N, \Sigma, P, S)$  then there exists a nonterminal  $A \in N$  such that  $A \rightarrow B \in P$  holds.
- ☐ Let  $A = (Q, \Sigma, \delta, q_0, F)$  be a deterministic finite automaton and let  $q \in Q$  be a state reachable from  $q_0$ . Then there exists a word  $w \in \Sigma^*$  with the property  $q_0 w \Rightarrow q$ .
- ☐ The class of regular languages is closed under the operation intersection.
- ☐ The class of context sensitive languages is closed under the operation union.
- ☐ Let  $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$  be a pushdown automaton. Then  $L(A)$  is of type 2.
- ☐ Let  $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$  be a pushdown automaton and let  $(x, r) \in \delta(z, q, b)$  be a transition of  $A$  ( $x, z \in Z, b \in \Sigma, q, r \in Q$ ). Applying this transition the stack will contain one more letter than in the previous configuration.
- ☐ Let  $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$  be a pushdown automaton. Then there is a grammar  $G$  in Chomsky normal form with the property  $L(G) = L(A)$ .
- ☐ Let  $G = (N, \Sigma, P, S)$  be a grammar in Chomsky normal form and consider a derivation tree  $t$  of a word from  $L(G)$ . Then all nodes of  $t$  have 0, 1 or 2 children.
- ☐ It is possible to prove that a given language  $L$  belongs to the class of context-free languages by applying Bar-Hillel Lemma for  $L$ .

2. Answer the following questions. No explanation required. (10x2 pont)

- (a) Give a formula describing the regular expression  $a^*b^*$ .

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- (b) What language operations are called regular operations?

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- (c) Let  $G = (N, \Sigma, P, S)$  be a grammar and let  $u, v \in (N \cup \Sigma)^*$ . Define what does it mean, that  $v$  is derivable from  $u$ . ( $u \Rightarrow_G^* v$ )

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- (d) Let  $A = (Q, \Sigma, \delta, Q_0, F)$  be a nondeterministic finite automaton. Define  $L(A)$ .

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- (e) Consider a nondeterministic finite automaton  $A = (\{p, q, r, s\}, \Sigma, \delta, \{p, r\}, \{p, q\})$  (i.e.,  $F = \{p, q\}$ ). List all the states in  $Q' \setminus F'$ , where  $A' = (Q', \Sigma, \delta', q'_0, F')$  is the deterministic finite automaton with  $Q' = P(Q)$  we have learnt to be equivalent with  $A$ .

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- (f) List all the suffix languages with respect to the the words over the alphabet  $\{a, b\}$  for the language  $L = \{a, bb\}$ .

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- (g) Assume, that the following sets are already known for an input of the CYK algorithm.

$$H_{22} = \{A, C\}, \quad H_{33} = \{B\}, \quad H_{44} = \{B\} \text{ and } H_{23} = \{B\}, \quad H_{34} = \{C\}.$$

Furthermore assume, that the rules of the grammar having only  $A, B$  and  $C$  on the right hand side are the following.

$$C \rightarrow CA \mid BC, \quad D \rightarrow CC \mid BB, \quad E \rightarrow BB.$$

Then  $H_{24} =$  .....

- (h) Give a pushdown automaton rule with the following effect. The automaton processes an input letter  $b$ , removes a letter  $c$  from the pusdown store and changes the state of the automaton from  $q_3$  to  $q_1$ .

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- (i) Place  $\mathcal{L}_{\text{detPDA}}$ , the class of languages being recognizable by a deteministic pushdown automaton in Chomsky's hierarchy of language classes.

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- (j) Define the transition function  $\delta$  of the pushdown automaton  $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$ .

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3. (3+4+7+6 points)
- (a) Let  $G = (N, \Sigma, P, S)$  be a grammar of type 2. Give the construction of a grammar of type 2 generating  $L(G)^*$  according to the construction we have learnt.
- (b) Define what is meant by a grammar in Chomsky normal form and state the normal form theorem with respect to it.
- (c) Present the length reduction algorithm we have learnt for regular grammars. Give the format of the rules before and after the length reduction.
- (d) Define what is meant by a suffix language with respect to a state  $q \in Q$  of a deterministic finite automaton  $A = (Q, \Sigma, \delta, q_0, F)$ .  
Give a system of linear language equations of size  $|Q|$ , where the unknowns are these suffix languages.